Fine-tuning ProGen2 on Green Fluorescent Protein (GFP)

The aim of this project was to fine-tune the ProGen2 language model on the green fluorescent protein (GFP) family and generate new sequence variants that could potentially retain the same fold and function. After training and inference, the designed sequences were evaluated using AlphaFold3, which provides confidence metrics about how reliable the predicted structures are.

Workflow Summary:

1. Model Fine-tuning
    - Pretrained model: progen2-small
    - Training was done on PSC Bridges2 GPU
2. Sequence Generation
    - Used the fine-tuned model to generate 92 new GFP variants.
3. Structure Prediction (AlphaFold3)
    - Each sequence was submitted to the AlphaFold3 web server.
    - For each job, AlphaFold produced model files (.cif) and JSONs containing confidence metrics.
    - A custom Python script was used to parse and rank all results based on their pLDDT and pTM values.

Alphafold3 metrics:

AlphaFold3 provides several confidence metrics that help interpret how reliable each predicted protein structure is. The first is pLDDT, which measures how confident the model is in the position of each residue. A higher pLDDT value means the atomic positions are predicted with greater certainty scores above 85 generally indicate very high confidence in the local structure. The second key metric is pTM (Predicted TM-score), which evaluates the confidence in the overall three-dimensional fold of the protein rather than individual residues.

Top-5 Candidates:

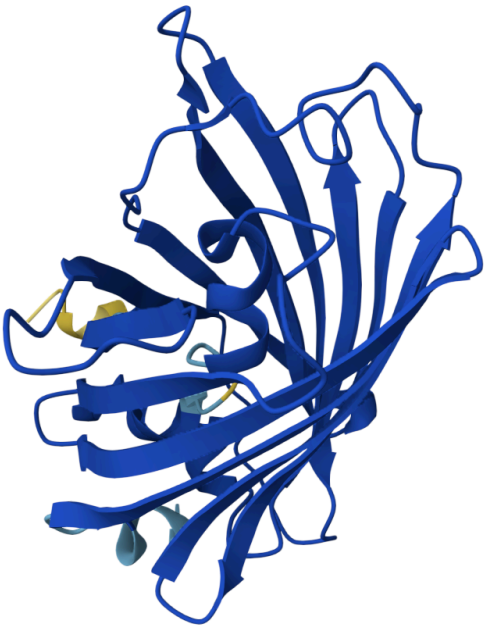
After evaluating all 92 predicted structures, the following five designs had the highest overall scores:

Sr. no	Design	pLDDT	pTM

1.	fold_design22	92.632	0.92
2.	fold_design92	92.455	0.92
3.	fold_design56	92.445	0.92
4.	fold_design18	92.336	0.92
5.	fold_design74	92.204	0.92

Representative Structures:

1. **Design 22**

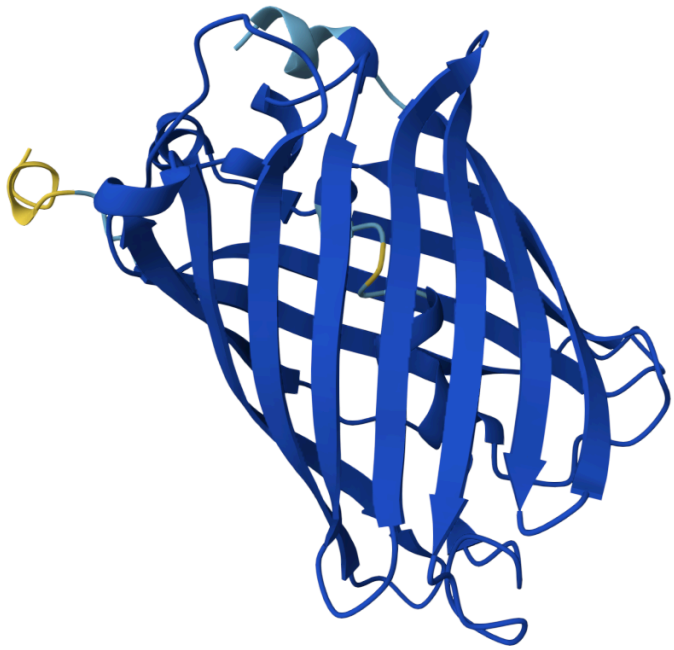


pLDDT: 92.632
pTM: 0.92
Sequence:
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDVAYGKLTLKFICTTGKLPVP
WPTLVTTLAYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRA
EVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVN

FKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHM
VLLEFVTAAGITHGMDELYK

This model shows an exceptionally well-formed β-barrel core with almost no disordered regions. The color gradient (blue regions indicating high confidence) is consistent across the entire structure, confirming a stable local geometry. With a global pTM of 0.92, this structure represents one of the most confidently folded candidates.

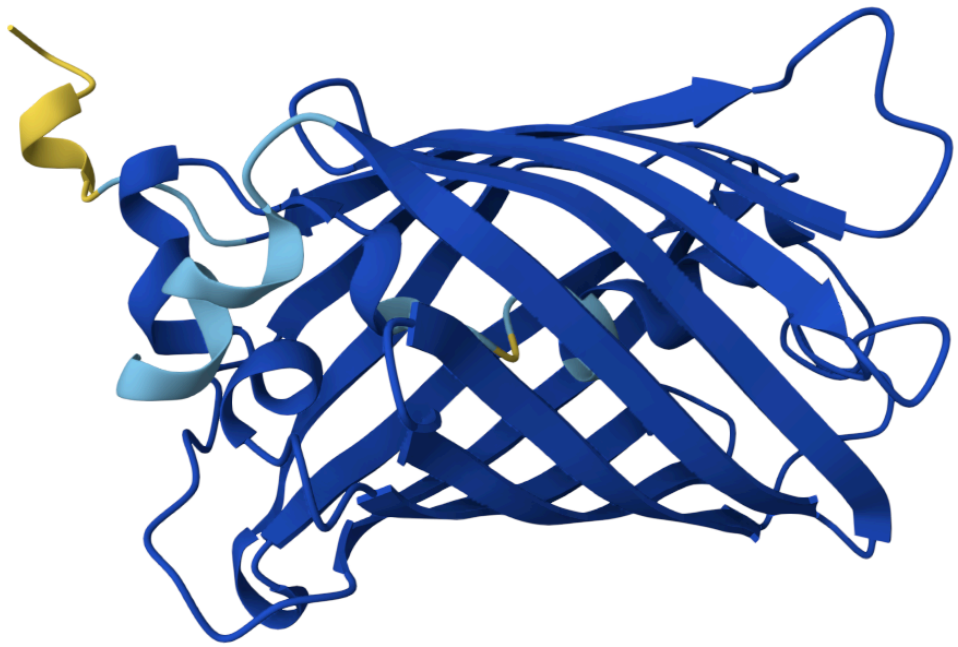
2. **Design 92:**



pLDDT: 92.455
iPTM: 0.92
Sequence:
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDVTYGKLTLKFICTTGKLPVP
WPTLVTTLSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRA
EVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVN
FKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHM
VLLEFVTAAGITHGMDELYK

This design maintains the overall GFP topology with an intact cylindrical β-barrel and short loop regions. The few yellow regions indicate mild flexibility rather than disorder. Its pTM value confirms excellent global folding, suggesting that the fine-tuned model successfully preserved essential GFP structural motifs.

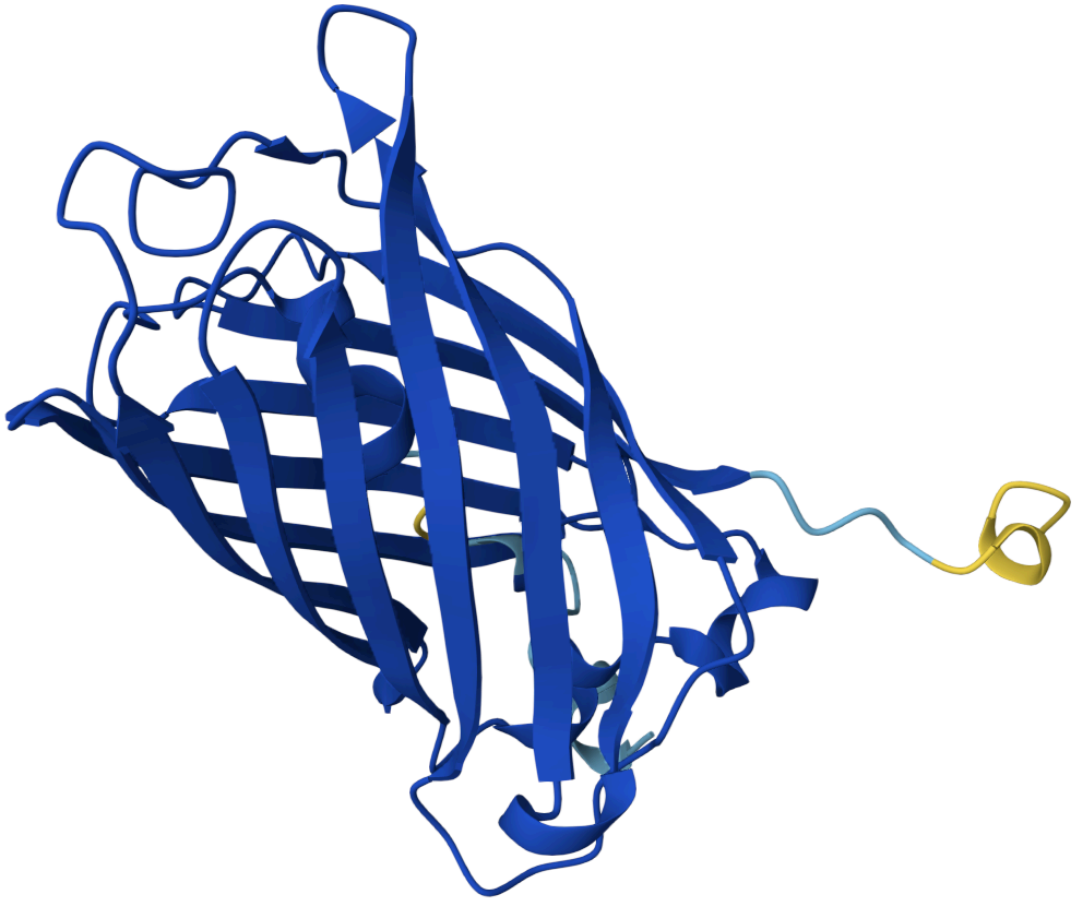
3. **Design 56**



pLDDT: 92.445
iPTM: 0.92
Sequence:
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDTTYGKLTLKFICTTGKLPVP
WPTLVTALSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRA
EVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVN
FKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHM
VLLEFVTAAGITHGMDELYK

fold_design56 features a clean β-barrel structure with high confidence throughout and well-defined secondary-structure elements. The minimal variation in pLDDT across the chain shows uniform model certainty. The barrel appears slightly more compact, which might indicate a stabilized internal packing or subtle residue substitutions predicted by ProGen2.

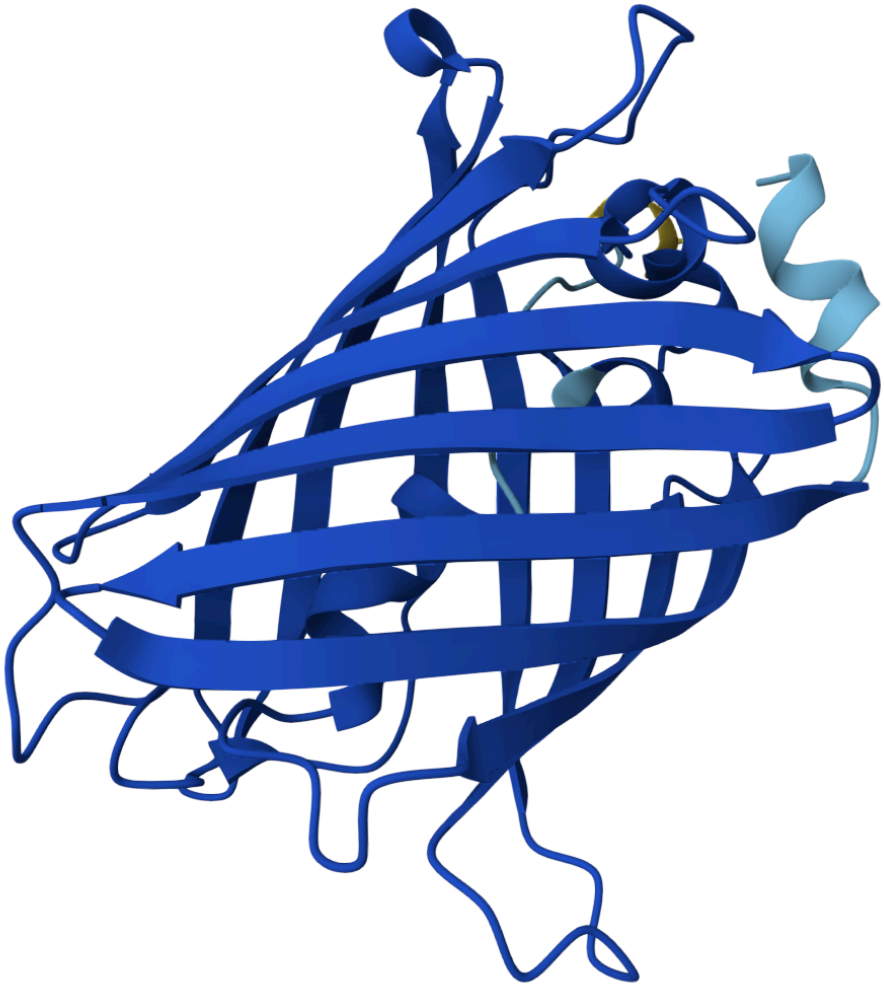
4. **Design 18**



pLDDT: 92.336
iPTM: 0.92
Sequence:
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDVTHGKLTLKFICTTGKLPVP
WPTLVTTLSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRA
EVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVN
FKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHM
VLLEFVTAAGITHGMDELYK

This variant displays the classical GFP fold with a small extended loop near the N-terminus. Despite this flexible loop, both pLDDT and pTM remain very high, indicating that the deviation does not destabilize the structure. The internal region appears well enclosed, implying good structural support for potential fluorescence.

5. **Design 74**



pLDDT: 92.204
iPTM: 0.92
Sequence:
SKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDTAYGKLTLKFICTTGKLPVP
WPTLVTTLSYGVQCFSRYPDHMKQHDFFKSAVPEGYVRERTIFFKDDGNYKTRA
EVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVN
FKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHM
VLLEFVTAAGITHGMDELYK

fold_design74 demonstrates consistent β-sheet alignment and minimal disorder. Its structure closely resembles the canonical GFP architecture, with stable hydrogen-bonding networks likely maintaining the barrel's integrity. The uniform blue coloring across the model reflects extremely reliable residue positioning.

Conclusion

Fine-tuning ProGen2 successfully adapted the model to the GFP family, producing multiple high-confidence variants. AlphaFold3 evaluation confirmed excellent fold stability (pLDDT > 90, pTM > 0.9). All five designs achieved pLDDT > 92 and pTM ≈ 0.92, confirming both strong local accuracy and excellent overall folding. The predicted structures are nearly identical in confidence and barrel geometry, suggesting the fine-tuning successfully constrained ProGen2 to the GFP family's structural domain. Subtle sequence differences likely influence surface flexibility rather than core stability.